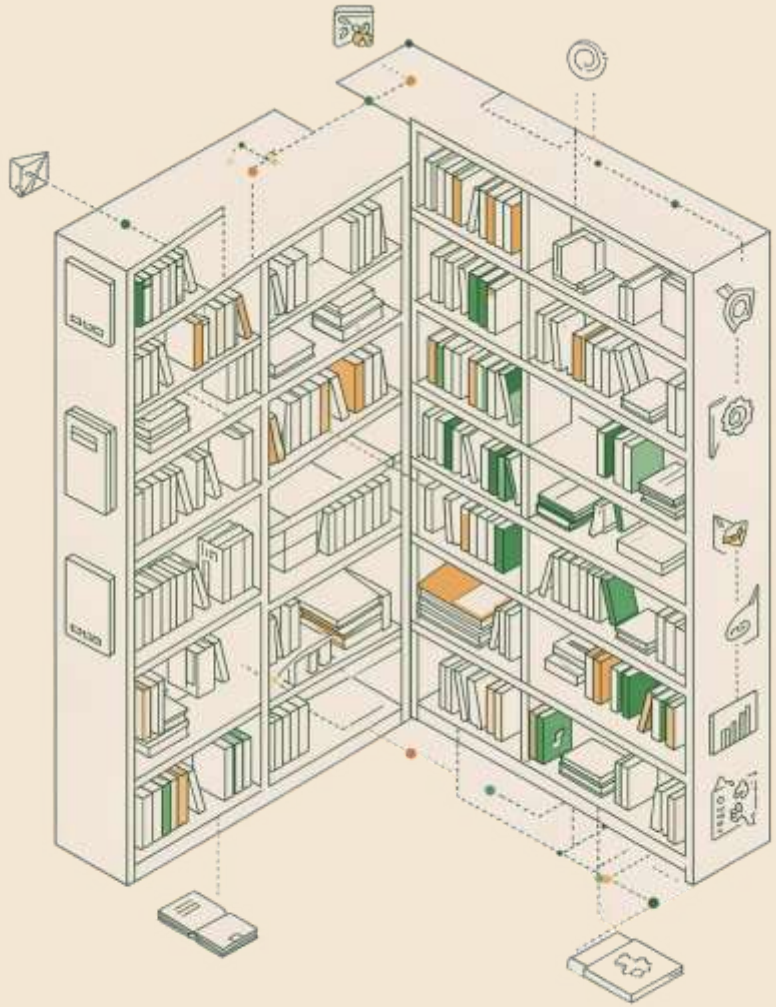
Library Digital Resource Management System

Design and Implementation of a Database

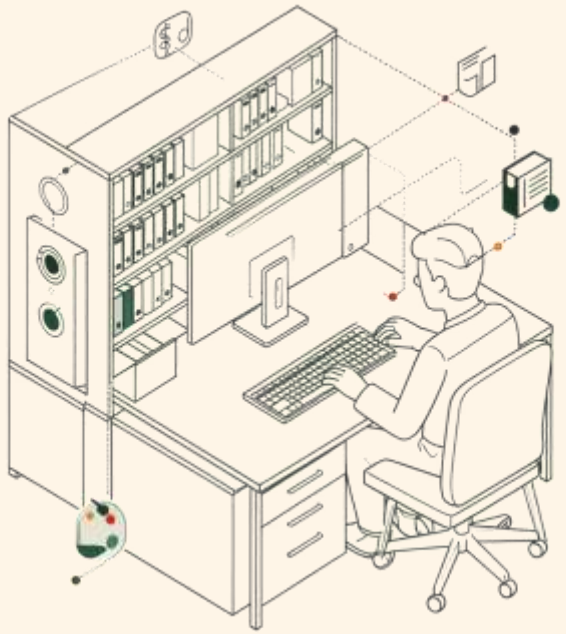




Part 1

Problem Identification & Requirements

Project Overview



What We're Building

A relational database system designed to power modern library operations — from physical shelves to digital archives.

The picture can't be displayed.

Physical Resources

Track books, editions, and individual copies across the entire collection

The picture can't be displayed.

Digital Access

Manage and log digital resource usage and access permissions

The picture can't be displayed.

Member Services

Enable comprehensive member management, reservations, and support

Problem Scope

Existing systems fall short in three critical areas that this project directly addresses:

Fragmented Inventory

No integrated tracking of books, editions, and individual physical copies — leading to confusion and errors

No Unified Member System

Member management and issue/return workflows exist in silos with no cohesive operational view

Disconnected Digital Layer

Digital resources require entirely separate access logging, leaving usage data unrecorded and unanalyzed

Key Objectives

1 Centralized Database

Consolidate all library resources and operations into a single, unified relational database

2 Inventory Tracking

Track resources across multiple editions and individual physical copies with full lifecycle visibility

3 Member Activity Management

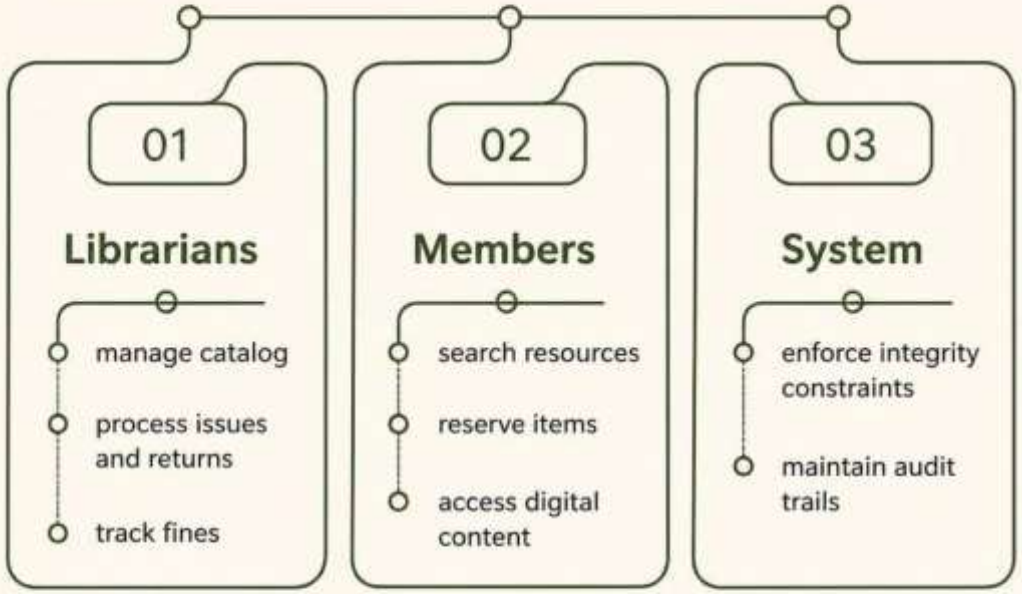
Manage reservations, issue/return workflows, and fine calculations for all registered members

4 Digital Access Logging

Record and audit every digital resource access event with member-level granularity



Primary Users & Functional Requirements



Who Uses the System

Librarians

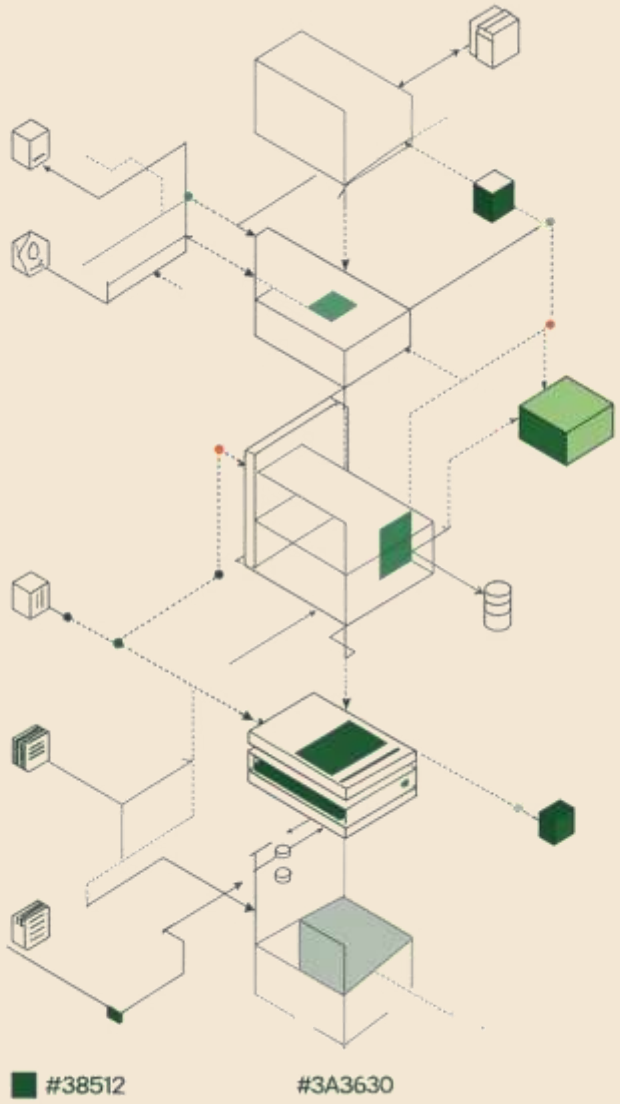
Manage the full catalog, process issue and return transactions, and track outstanding fines

Members

Search available resources, place reservations on items, and access digital content

System

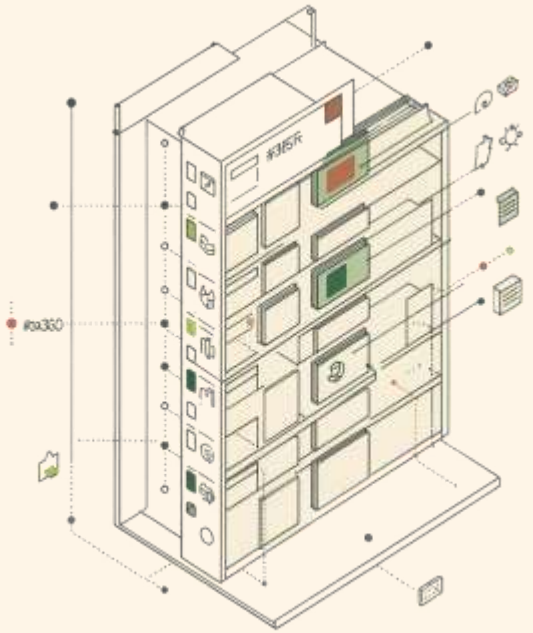
Enforce referential integrity constraints and maintain complete audit trails automatically



Part 2

Conceptual Design & ER Diagram

Principal Entities



Resource & Inventory Layer

The core entities that define what the library holds and how it is organized:

Books
Core bibliographic records

Editions
Versioned publication records per book

Authors
Creator attribution records

Publishers
Publishing house records

Categories
Genre and subject classifications

Copies
Individual physical items on shelf

Entity Relationships



How Entities Connect

Resource Hierarchy

Authors write **Books**; Books have multiple **Editions**; **Publishers** produce Editions; Editions contain multiple **Copies**

Member Transactions

Members issue **Copies** and create **Reservations** on unavailable items

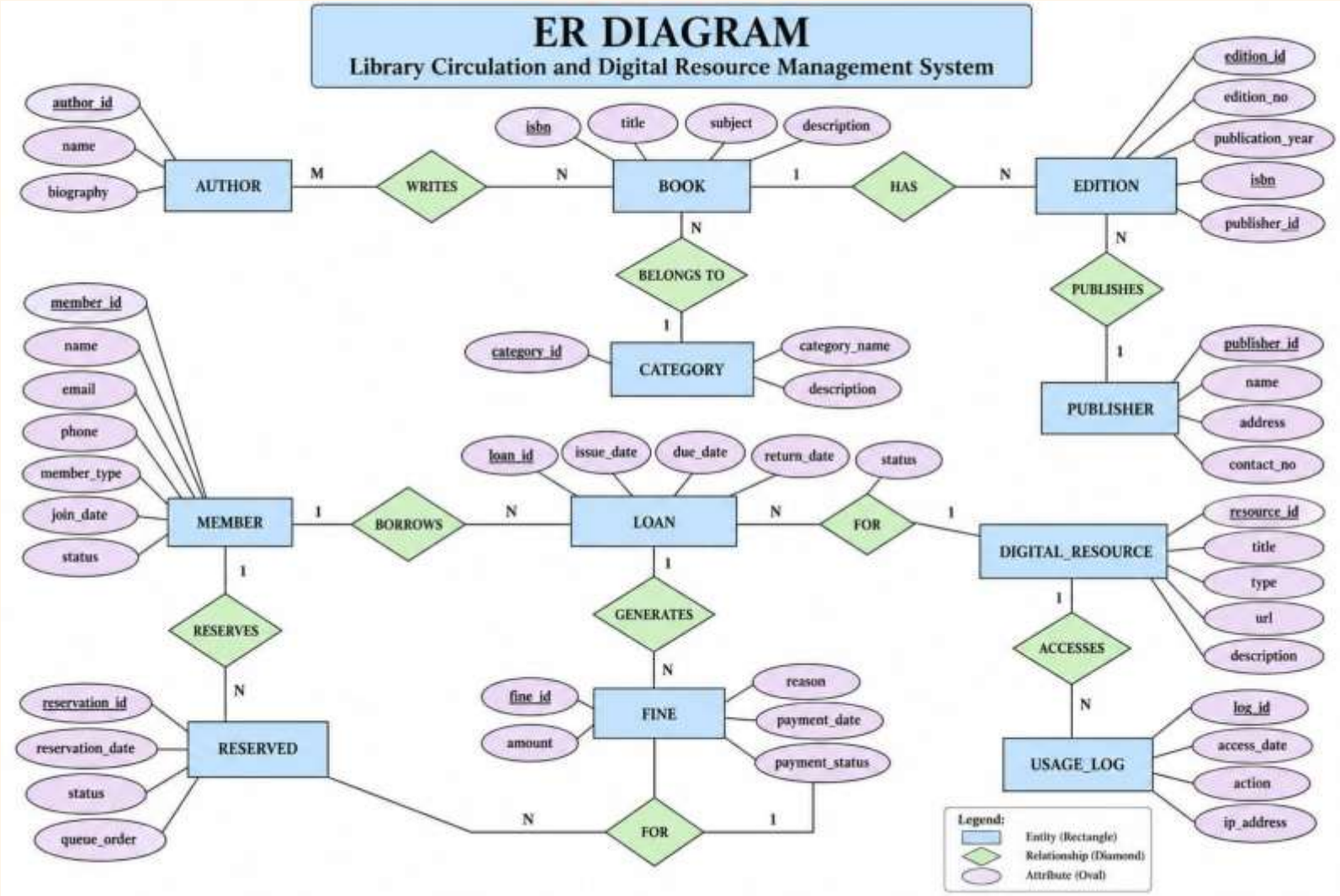
Fine Generation

Issues generate **Returns**; overdue items automatically trigger **Fines**

Digital Tracking

Digital-Access Logs capture every Member interaction with digital resources

Formal ER Diagram (Chen's Notation)



Relational Schema

This section outlines the comprehensive relational schema for the library digital resource management system. It details all tables, their column definitions, inferred data types, and primary/foreign key relationships.

Data types have been inferred based on standard database practices and the nature of the data.

Core Resource Tables

Books

Column Name	Data Type	Keys/Relationships
BookID	INT	PK
Title	VARCHAR(255)	
ISBN	VARCHAR(13)	
CategoryID	INT	FK (References Categories.CategoryID)

Editions

Column Name	Data Type	Keys/Relationships
EditionID	INT	PK
BookID	INT	FK (References Books.BookID)
PublisherID	INT	FK (References Publishers.PublisherID)
Year	YEAR	
Format	VARCHAR(50)	

Copies

Column Name	Data Type	Keys/Relationships
CopyID	INT	PK
EditionID	INT	FK (References Editions.EditionID)
Status	VARCHAR(50)	
AcquisitionDate	DATE	

Authors

Column Name	Data Type	Keys/Relationships
AuthorID	INT	PK
Name	VARCHAR(255)	
Biography	TEXT	
DateOfBirth	DATE	